eGamma-VBA: expanded Gamma distribution for Microsoft Excel

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Frode Drevland

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CHAPTER

ONE

ABOUT

1.1 Introduction

The expanded gamma distribution extends the traditional three-parameter gamma distribution to also accommodate left-skewed data. Its versatility is particularly beneficial in project cost estimation and other fields where such data patterns are common. Although the distribution has been in de facto use in Scandinavian cost-estimation practice since the 1970s, it has only recently been formally defined. The *EGAMMA-VBA* library provides a Python implementation of the expanded gamma distribution, along with the tools for its practical application.

The *EGAMMA-VBA* library enhances builds on implementation of the gamma distribution, enhancing it to incorporate the specialized attributes of the expanded version. Additionally, *EGAMMA-VBA* introduces capabilities beyond those offered by SciPy, such as three-point estimation. This method, used with expert judgment, determines the most likely, optimistic, and pessimistic scenarios. Unlike standard practice, which often places optimistic and pessimistic estimates at the 10th and 90th percentiles, *EGAMMA-VBA* offers the flexibility to choose any percentile range, allowing for customized three-point estimation according to user preference.

1.2 Documentation Content

- **About:** The current section.
- **Mathematical Definitions:** Mathematical underpinnings of the distribution's functions and statistical measures.
- **Installation:** Instructions for installing the library.
- **Usage:** Introduction to the library's basic usage.
- **Functions:** In-depth descriptions of the library's functions, including parameters and return values.

1.3 Support

If you encounter any problems or have any questions, please open an issue on the GitHub repository issue tracker.

1.4 Contact Information

For inquiries or collaboration on the *EGAMMA-VBA* library, contact:

- **Associate Professor Frode Drevland**
- **Affiliation**: Norwegian University of Science and Technology (NTNU)
- **Email**: frode.drevland@ntnu.no

Dr. Drevland is dedicated to the continuous development of the *EGAMMA-VBA* library and welcomes feedback, suggestions, and contributions from the community.

1.5 License

This library is distributed under the MIT License. See LICENSE for more information.

CHAPTER

TWO

MATHEMATICAL DEFINITIONS

2.1 Introduction

This page is the reference for the mathematics behind the library: the density and its cumulative distribution function, the statistical measures, and the procedure by which the parameters are recovered from a three-point estimate. Nothing here is needed to use the library; see *Usage* for that.

2.2 Parameters

The distribution is characterised by three parameters:

- α (alpha): The shape parameter, which must be positive ($\alpha > 0$).
- β (beta): The scale parameter, which must be non-zero ($\beta \neq 0$). Its sign determines the direction of skewness: positive values give right skew, negative values left skew.
- δ (delta): The location parameter, which may be any real number. It is the bound of the support, lower when $\beta > 0$ and upper when $\beta < 0$.

The support is $[\delta, \infty)$ for a positive scale and $(-\infty, \delta]$ for a negative one.

The distribution is strictly skewed in one direction or the other; exact symmetry is not attainable at any finite α. As α grows the shape converges to that of a normal distribution, so a large value such as 10^9 approximates a symmetric case closely.

2.3 Distribution Functions

The probability density function (PDF) and cumulative distribution function (CDF) are defined as follows.

Probability Density Function (PDF):

$$f(x; \alpha, \beta, \delta) = \begin{cases} \dfrac{|x-\delta|^{\alpha-1} e^{-(x-\delta)/\beta}}{|\beta|^{\alpha}\Gamma(\alpha)}, & \text{for } \dfrac{x-\delta}{\beta} > 0 \\ 0, & \text{otherwise} \end{cases}$$

where $\Gamma(\alpha)$ is the gamma function $\displaystyle\int_0^\infty t^{\alpha-1} e^{-t}\, dt$

Cumulative Density Function (CDF):

$$F(x;\alpha,\beta,\delta) = \begin{cases} \dfrac{\gamma(\alpha, \dfrac{x-\delta}{\beta})}{\Gamma(\alpha)}, & \text{for } \beta > 0, \dfrac{x-\delta}{\beta} > 0 \\ 1 - \dfrac{\gamma(\alpha, \dfrac{x-\delta}{\beta})}{\Gamma(\alpha)}, & \text{for } \beta < 0, \dfrac{x-\delta}{\beta} > 0 \end{cases}$$

where $\gamma\left(\alpha, \frac{x-\delta}{\beta}\right)$ is the lower incomplete gamma function $\int_0^{\frac{x-\delta}{\beta}} t^{\alpha-1}e^{-t}\,dt$.

The CDF has no closed form, so the percent point function must be obtained numerically. That single fact shapes the fitting procedure below.

2.4 Statistical measures

Expected value:

$$E[X] = \alpha\beta + \delta$$

Variance:

$$\mathrm{Var}(X) = \alpha\beta^2$$

Skewness:

$$\mathrm{Skew}(X) = \frac{\beta}{|\beta|} \times \frac{2}{\sqrt{\alpha}}$$

Excess kurtosis:

$$\mathrm{ExKurt}(X) = \frac{6}{\alpha}$$

Mode, for $\alpha > 1$:

$$\mathrm{Mode}(X) = (\alpha - 1)\beta + \delta$$

For $\alpha \leq 1$ the density is monotone on its support and the mode lies at δ.

2.5 Fitting to a three-point estimate

Given elicited values $L \leq M \leq H$, where L and H are the P_L and $P_H = 1 - P_L$ percentiles and M is the mode, the problem is to find α, β and δ reproducing all three.

Writing $G^{-1}(p;\alpha)$ for the percent point function of the standard gamma distribution with unit scale and zero location, and taking $\beta > 0$ for the moment, the ratio of the two half-ranges eliminates both the scale and the location and leaves a single equation in the shape parameter:

$$\frac{M - L}{H - M} = \frac{(\alpha - 1) - G^{-1}(P_L;\alpha)}{G^{-1}(P_H;\alpha) - (\alpha - 1)}$$

This has no closed-form solution and is solved numerically. The library bisects on skewness, $2/\sqrt{\alpha}$, rather than on α itself, since skewness is bounded on $(0, 2]$ and needs no interval derived from the ceiling.

Excel's `GAMMA.INV` is the numerical foundation for all of this, and its accuracy at very large shape parameters is the practical reason the ceiling is set where it is.

With α known, the remaining parameters follow in closed form. The scale comes from the full elicited span, which is better conditioned than either half:

$$\beta = \frac{H - L}{G^{-1}(P_H; \alpha) - G^{-1}(P_L; \alpha)}$$

and the location from the mode relation:

$$\delta = M - (\alpha - 1)\beta$$

A left-skewed estimate, one for which $M - L > H - M$, is the mirror image. It is solved by inverting the ratio, solving the same equation, and negating the scale before computing the location.

2.6 Admissible estimates

Not every ordering $L \leq M \leq H$ can be fitted. The mode of the fitted distribution cannot fall outside the interval the elicited percentiles enclose, and requiring $L \leq M$ therefore requires $G^{-1}(P_L; \alpha) \leq \alpha - 1$. The shape at which the two coincide is the smallest admissible one. For $P_L = 0.10$ it is $\alpha = 1.15631$, corresponding to a skewness of 1.85991, so no valid estimate implies more skewness than that. The bound depends on P_L; a wider elicited interval admits more.

The boundary at $\alpha = 1$, where the mode reaches the support endpoint, lies strictly inside the inadmissible region and is never reached from a valid estimate.

A symmetric estimate is reproducible only in the limit $\alpha \to \infty$. The library returns a ceiling of 10^9, which reproduces the elicited values to about 1.5×10^{-5} of the elicited range rather than exactly.

2.7 Accuracy

For any admissible asymmetric estimate, the deviation of a recovered elicited value, normalised by the elicited range, is bounded by

$$E < \frac{\tau}{2(2 - \tau)} = \frac{\tau}{4} + O(\tau^2)$$

where τ is the relative acceptance threshold on the ratio. At the default $\tau = 10^{-10}$ this is 2.5×10^{-11}. The bound holds for every admissible input rather than for tested cases, and it excludes the symmetric branch, where the ceiling rather than the threshold governs.

Recovery of α itself is a weaker guarantee. The ratio above approaches unity as $\alpha^{-1/2}$, so the recovery becomes increasingly ill-conditioned near symmetry and a fixed threshold buys progressively less accuracy in the shape parameter. This is a property of the problem and not of the solver; tightening the threshold improves the recovered shape at the cost of more iterations.

CHAPTER

THREE

INSTALLATION

To integrate the EGAMMA library into Excel, you have two distinct installation methods to choose from, depending on your needs and preferences.

Installing as an .xlam Add-in for Global Access If you require the EGAMMA library to be accessible across all Excel workbooks on your system, the optimal approach is to install it as an .xlam add-in. This method ensures that the library's features are readily available in Excel, no matter which workbook you open.

Importing VBA Code for Workbook-Specific Use Alternatively, if your intention is to utilize the EGAMMA library in only a few selected workbooks, you may opt to import the library's code directly into those workbooks via Excel's VBA code editor. This approach allows for a more targeted use of the library, without affecting the global Excel environment.

For comprehensive guidance on executing these installation methods, please refer to the following instructional pages, which offer step-by-step procedures tailored to each approach.

3.1 Install the EGAMMA.xlam Plugin for Excel

This page provides comprehensive instructions for installing the EGAMMA.xlam plugin, enabling the use of EGAMMA functions universally across all Excel workbooks. For a more selective approach, you have the option to import the EGAMMA code into individual workbooks as needed (refer to *Importing the EGAMMA VBA Code into an Excel Workbook* for details).

To equip your Excel environment with the EGAMMA.xlam plugin, please proceed with the following download and installation steps:

3.1.1 Step 1: Download the Plugin

- Visit the [EGAMMA-VBA GitHub Releases page](https://github.com/FrodeDrevland/EGAMMA-VBA/releases/latest)
- Click on the download link for the EGAMMA.xlam file.
- Save the file to a known location on your computer, such as your Desktop or a dedicated Downloads folder.

3.1.2 Step 2: Open Excel

- Launch Microsoft Excel on your computer.

3.1.3 Step 3: Access the Add-ins Menu

- For Excel 2010 and later: - Go to the `File` tab, click on `Options`, and then select `Add-Ins`. - At the bottom of the window, in the `Manage` box, select `Excel Add-ins` and click `Go`.
- For Excel 2007: - Click the Microsoft Office Button, then click `Excel Options`, and choose `Add-Ins`. - At the bottom of the window, in the `Manage` box, select `Excel Add-ins` and click `Go`.

3.1.4 Step 4: Install the Plugin

- In the Add-Ins window, click `Browse...`.
- Navigate to the location where you saved the EGAMMA.xlam file.
- Select the file and click `OK`.
- Ensure that the checkbox next to EGAMMA is checked in the list of available Add-Ins.
- Click `OK` to close the Add-Ins window.

3.1.5 Step 5: Confirm Installation

- The EGAMMA plugin should now be installed and ready to use.
- Confirm the installation by looking for by checking that the EGAMMA functions work (See *Usage* documentation).

3.1.6 Step 6: Enable Macros if Necessary

- The plugin relies on includes macros. If the formuals are not working, you may need to enable macros: - Go to the `File` tab, click on `Options`, then select `Trust Center`, and click on the `Trust Center Settings...` button. - In the Trust Center, select `Macro Settings` and choose the appropriate setting to enable macros. Be cautious and only enable macros from trusted sources.

3.1.7 Troubleshooting

- If the EGAMMA functions does not appear, restart Excel and try again.
- If you encounter any issues, check that your version of Excel supports .xlam plugins and that you have the necessary permissions to install add-ins on your system.

3.2 Importing the EGAMMA VBA Code into an Excel Workbook

This guide outlines the steps to integrate the EGAMMA VBA code into an individual Excel workbook, which allows you to utilize EGAMMA's capabilities within that specific file. If you wish to extend EGAMMA's functionalities to all workbooks on your system, consider installing the EGAMMA.xlam plugin instead (see *Install the EGAMMA.xlam Plugin for Excel* for installation guidance).

To begin using the EGAMMA VBA code within a single workbook, you will need to download the EGAMMA.bas file. This file is a VBA module containing the necessary code to be imported into the VBA editor of Excel. Doing so enables you to tailor and engage with the EGAMMA features directly in your chosen Excel workbook.

Please follow the detailed steps below to import the EGAMMA.bas file into your workbook:

3.2.1 Step 1: Download the EGAMMA.bas File

- Visit the [EGAMMA-VBA GitHub Releases page](https://github.com/FrodeDrevland/EGAMMA-VBA/releases/latest) to access the latest version of the EGAMMA.bas file.
- Click on the link for the EGAMMA.bas file to initiate the download.
- Save the file to a familiar directory on your computer for easy retrieval.

3.2.2 Step 2: Enable the Developer Tab in Excel

- For Excel 2010 and newer: - Navigate to the `File` tab, select `Options`, and then choose `Customize Ribbon`. - In the customization area, ensure the `Developer` checkbox is selected and confirm with `OK`.
- For Excel 2007: - Click the Microsoft Office Button, select `Excel Options`, and within the `Popular` category, activate the `Show Developer tab in the Ribbon` option.

3.2.3 Step 3: Access the VBA Editor

- From the Developer tab, initiate the `Visual Basic` command to open the VBA editor.

3.2.4 Step 4: Import the .bas File

- Within the VBA editor, right-click on the `VBAProject` pertaining to your workbook.
- Choose `Import File...` and navigate to the directory where the EGAMMA.bas file is located.
- Select the file and opt for `Open` to import it into your project.

3.2.5 Step 5: Save Your Workbook with Macros

- Return to Excel and save your workbook by selecting `Save`.
- If your workbook is in .xlsx format, you'll need to switch to the .xlsm format to preserve the VBA code.

3.2.6 Step 6: Enable Macros if Necessary

- To execute the VBA code, ensure that macros are enabled by visiting the `File` tab, selecting `Options`, followed by `Trust Center`, and then `Trust Center Settings...`.
- In the Macro Settings section, choose a suitable option to enable macros, keeping security in mind.

3.2.7 Troubleshooting Tips

- If the Developer tab or VBA editor is not visible, they may need to be enabled in your Excel settings.
- Should you face any issues with the code execution, verify the compatibility of the code with your Excel version and check for required references.

Always back up your data before importing a .bas file, particularly if it includes extensive macros or code that could modify your data significantly.

CHAPTER

FOUR

USAGE

Once the EGAMMA plugin is installed, its functions become accessible within your Excel workbook(s). This section offers a general overview of how to use these functions.

4.1 EGAMMA_DIST and EGAMMA_INV Functions

The EGAMMA_DIST and EGAMMA_INV functions are analogous to Excel's native GAMMA.DIST and GAMMA.INV functions, respectively. The key enhancements in the EGAMMA versions include an additional *delta* parameter for location adjustment, and the ability for the *beta* parameter to accept negative values, accommodating negatively skewed distributions.

For instance, to calculate the cumulative probability for a value of 200 in an expanded distribution characterized by alpha = 10, beta = 20, and delta = 100, use:

```
=EGAMMA_DIST(200, 10, 20, 100, TRUE)
```

4.2 Statistical Measures

The EGAMMA library includes a range of functions designed to compute various statistical measures based on the parameters of the expanded gamma distribution (alpha, beta, and delta). These functions operate similarly to standard Excel functions:

- `EGAMMA_MEAN(alpha, beta, delta)`: Calculates the mean.
- `EGAMMA_MODE(alpha, beta, delta)`: Calculates the mode.
- `EGAMMA_MEDIAN(alpha, beta, delta)`: Calculates the median.
- `EGAMMA_VAR(alpha, beta)`: Calculates the variance.
- `EGAMMA_STDDEV(alpha, beta)`: Calculates the standard deviation.
- `EGAMMA_SKEW(alpha, beta)`: Calculates the skewness.
- `EGAMMA_KURT(alpha)`: Calculates the excess kurtosis.

These are functions of the distribution's parameters alone, and apply equally whether the parameters came from a three-point estimate, from a fit to data, or were entered directly.

4.3 Parameter estimation

The EGAMMA library offers two distinct functions for parameter estimation: EGAMMA_TPE_TO_PARAMS and EGAMMA_FIT_TO_PARAMS. These functions are designed to calculate the parameters of the Expanded Gamma Distribution either from a three-point estimate or by fitting to a data set.

4.3.1 EGAMMA_TPE_TO_PARAMS

The EGAMMA_TPE_TO_PARAMS function requires three key values as arguments: the low, most likely, and high values of a three-point estimate. Additionally, it accepts an optional low probability value, which is the probability of the low estimate (defaulting to 0.1). It's important to note that the probability of the high estimate is always the complement of the low probability (1 - low probability).

```
=EGAMMA_TPE_TO_PARAMS(low, most-likely, high, [low-probability])
Example: =EGAMMA_TPE_TO_PARAMS(100, 200, 400, 0.1)
```

The function outputs a 3x1 array: alpha in the function's cell, and beta and delta in the two adjacent cells to the right. If these adjacent cells are not empty, the function will return `#SPILL!`, indicating it cannot display the results.

The fitted parameters recover the three elicited values to within 2.5×10^{-11} of the elicited range. The exception is an estimate too close to symmetry to resolve: a perfectly symmetric estimate corresponds to an infinite shape parameter, so those are fitted at the library's finite ceiling and recover the elicited values to about 1.5×10^{-5} of the range instead. Such a fit returns a shape parameter of exactly 10^9, the library's ceiling, which is how you recognise the case; no other estimate produces that value.

A most likely value that equals, or nearly equals, one of the outer values is admissible and is fitted like any other estimate; the endpoint case is not treated separately. An estimate is rejected with `#N/A` only if the three values are not ordered, that is unless `low <= most-likely <= high` and `low < high`.

4.3.2 EGAMMA_FIT_TO_PARAMS

EGAMMA_FIT_TO_PARAMS is designed to estimate the alpha, beta, and delta parameters of the Expanded Gamma Distribution from a given data set. This function uses the method of moments, employing the skewness of the data to estimate alpha, the standard deviation for beta, and the mean, alpha, and beta to calculate delta.

```
=EGAMMA_FIT_TO_PARAMS(range)
Example: =EGAMMA_FIT_TO_PARAMS(A1:A100)
```

Similar to EGAMMA_TPE_TO_PARAMS, the results are returned as an array. The alpha parameter appears in the cell where the function is entered, with beta and delta in the adjacent right cells.

For further details, consult the *Functions* documentation.

CHAPTER

FIVE

FUNCTIONS

EGAMMA_DIST(x, alpha, beta, delta, cumulative):

Calculate the expanded gamma distribution.

Parameters

- **x** – Required. The value at which the function is evaluated.
- **alpha** – Required. The shape parameter of the distribution.
- **beta** – Required. The scale parameter of the distribution. Negative beta entails a left skewed distribution.
- **delta** – Required. The location parameter of the distribution.
- **cumulative** – Required. A logical value that determines the form of the function. If cumulative is TRUE, GAMMA.DIST returns the cumulative distribution function; if FALSE, it returns the probability density function.

Returns

The value of the (cumulative) gamma distribution.

Return type

float

EGAMMA_INV(probability, alpha, beta, delta):

Calculate the inverse of the expanded gamma distribution.

Parameters

- **probability** – The probability for which the inverse is calculated.
- **alpha** – The shape parameter of the expanded gamma distribution.
- **beta** – The scale parameter of the expanded gamma distribution.
- **delta** – The location parameter of the expanded gamma distribution.

Returns

The inverse value of the expanded gamma distribution for the given probability.

EGAMMA_MEAN(alpha, beta, delta):

Calculate the mean of the expanded gamma distribution.

Parameters

- **alpha** – The shape parameter of the expanded gamma distribution.
- **beta** – The scale parameter of the expanded gamma distribution.

- **delta** – The location parameter of the expanded gamma distribution.

Returns
The mean of the expanded gamma distribution.

EGAMMA_MODE(alpha, beta, delta):

Calculate the mode of the expanded gamma distribution.

For $\alpha > 1$ the density has an interior maximum at $(\alpha - 1)\beta + \delta$. For $0 < \alpha \leq 1$ it is monotone on its support and the mode lies at the support boundary, δ; the interior expression would place it outside the support altogether. A three-point fit never returns a shape at or below 1, so this distinction arises only for parameters entered directly.

Parameters

- **alpha** – The shape parameter of the expanded gamma distribution.
- **beta** – The scale parameter of the expanded gamma distribution.
- **delta** – The location parameter of the expanded gamma distribution.

Returns
The mode of the expanded gamma distribution, or `#NUM!` if the shape parameter is not positive.

EGAMMA_MEDIAN(alpha, beta, delta):

Calculate the median of the expanded gamma distribution.

Parameters

- **alpha** – The shape parameter of the expanded gamma distribution.
- **beta** – The scale parameter of the expanded gamma distribution.
- **delta** – The location parameter of the expanded gamma distribution.

Returns
The median of the expanded gamma distribution.

EGAMMA_VAR(alpha, beta):

Calculate the variance of the expanded gamma distribution.

Parameters

- **alpha** – The shape parameter of the expanded gamma distribution.
- **beta** – The scale parameter of the expanded gamma distribution.

Returns
The variance of the expanded gamma distribution.

EGAMMA_STDDEV(alpha, beta):

Calculate the standard deviation of the expanded gamma distribution.

Parameters

- **alpha** – The shape parameter of the expanded gamma distribution.
- **beta** – The scale parameter of the expanded gamma distribution.

Returns
The standard deviation of the expanded gamma distribution.

EGAMMA_SKEW(alpha, beta):

Calculate the skewness of the expanded gamma distribution.

Parameters

- **alpha** – The shape parameter of the expanded gamma distribution.
- **beta** – The scale parameter of the expanded gamma distribution.

Returns

The skewness of the expanded gamma distribution.

EGAMMA_KURT(alpha):

Calculate the excess kurtosis of the expanded gamma distribution.

Parameters

alpha – The shape parameter of the expanded gamma distribution.

Returns

The excess kurtosis of the expanded gamma distribution.

EGAMMA_FIT_TO_PARAMS(*args):

Fit the parameters of the expanded gamma distribution to a given set of data points.

This function estimates the alpha, beta, and delta parameters of the expanded gamma distribution, based on the provided data using the method of moments. It uses the skewness of the data to estimate alpha, then calculates beta using the standard deviation, and finally computes delta from the mean, alpha, and beta.

Parameters

args – A variable number of arrays or lists containing numerical data points. These are combined into a single dataset for the analysis.

Returns

A list containing the estimated parameters [alpha, beta, delta] of the extended gamma distribution.

Raises

ValueError – If the input data does not allow for a proper fit, e.g., if it leads to divide-by-zero errors.

EGAMMA_TPE_TO_PARAMS(low, likely, high, low_probability=0.1, fit_tolerance=2.5E-11, ceiling_tolerance=2E-5, require_nonnegative=FALSE):

Estimate the parameters of the expanded gamma distribution based on a three-point estimate.

This function uses a low, likely, and high estimate to determine the parameters of the extended gamma distribution.

The shape parameter is found by bisection on the magnitude of the skewness. The quantity matched is the mode's position within the elicited range,

$$t = \frac{\min(\text{likely} - \text{low},\ \text{high} - \text{likely})}{\text{high} - \text{low}},$$

against the position the standard Gamma with that shape produces between its two percentiles,

$$t_\alpha = \frac{\alpha - 1 - q_L}{q_H - q_L},$$

and the search stops when the two differ by less than 2.5×10^{-11}. That difference is itself the largest error, as a fraction of the elicited range, with which the three elicited values are recovered from the fitted parameters, so the tolerance is stated directly in the quantity of interest. A most likely value equal to an outer value gives $t = 0$ and goes through the same search; there is no separate case for it.

An estimate too near symmetry to resolve cannot be fitted to that tolerance, because a perfectly symmetric estimate is only reproducible in the limit as the shape tends to infinity. Such an estimate is fitted at the library's finite ceiling instead, and the returned shape is then exactly 10^9, which is how a caller recognises the case. The elicited values are reproduced to about 1.5×10^{-5} of the elicited range rather than to the tolerance, and `EGAMMA_TPE_DIAGNOSTICS()` reports the case as `"approximation"`.

Once the parameters are found, the result is accepted only if three rules hold. The tolerance is `fit_tolerance` for an ordinary fit and `ceiling_tolerance` for a fit at the ceiling.

1. The position residual $|t_\alpha - t|$, which is the normalised reproduction error in exact arithmetic, is within the tolerance. No allowance is added.
2. The estimated rounding uncertainty of the reconstruction check, $4\,\epsilon_{\text{mach}}\,m/(\text{high} - \text{low})$, where m is the largest magnitude among the elicited values, the location and the scaled quantiles, is no larger than the tolerance. Otherwise the check cannot verify the accuracy asked for: in double precision this happens at the default tolerance once the values exceed about 28,000 times their range. The factor 4 is an estimate from a first-order count of rounded operations, not a proven bound.
3. The low, likely and high values reconstructed from the parameters agree with those supplied to within the tolerance plus that uncertainty.

A result that fails any rule returns `#NUM!`, and `EGAMMA_TPE_DIAGNOSTICS()` names the rule.

Parameters

- **`low`** – The low estimate of the distribution.
- **`likely`** – The most likely estimate of the distribution.
- **`high`** – The high estimate of the distribution.
- **`low_probability`** – The probability of the low estimates (default is 0.1).
- **`fit_tolerance`** – The position residual accepted for an ordinary fit, as a fraction of the range (default 2.5E-11). Loosen it for an estimate refused as unverifiable, or fit shifted values instead.
- **`ceiling_tolerance`** – The position residual accepted at the shape ceiling, as a fraction of the range (default 2E-5). All five supported percentile conventions meet the default for a symmetric estimate; a convention near the median, such as 0.4999, does not.
- **`require_nonnegative`** – TRUE to refuse, with `#NUM!`, a fit whose distribution has support below zero (default FALSE).

Returns

A list containing the estimated parameters [alpha, beta, delta] of the extended gamma distribution.

Raises

`ValueError` – Returns `#N/A` if the input values are inconsistent (low > likely, high

< likely, or low >= high), and `#NUM!` if an argument is out of range or the fit fails any of the three rules or the non-negative support requirement.

EGAMMA_TPE_DIAGNOSTICS(low, likely, high, low_probability=0.1, fit_tolerance=2.5E-11, ceiling_tolerance=2E-5, require_nonnegative=FALSE):

Report on the fit `EGAMMA_TPE_TO_PARAMS()` makes for the same arguments, including why it failed.

Returns

A list of five values. First the status: `"converged"` or `"approximation"` (a fit at the shape ceiling) for an accepted fit, and otherwise `"ceiling tolerance not met"`, `"precision insufficient"`, `"reconstruction inconsistent"`, `"nonnegative support not met"`, `"search failed"` or `"numerical failure"`. Then the position residual, the reconstructed error as computed, and the verification uncertainty, each as a fraction of the range; and TRUE if the fitted distribution has support below zero. That is always so for a negative scale, whose support is unbounded below, and for a positive scale with a negative location. Values not computed before a failure are `#N/A`.

Raises

ValueError – Returns the same error value as `EGAMMA_TPE_TO_PARAMS()` when an argument is invalid.